

ABHIYAN SAINJU

Washington, DC

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Education

George Washington University

MS Computer Science — **GPA: 3.97/4.0** — **SEAS Dean's Award Recipient**
AWS Certifications: Cloud Operations (Nov 2024), Cloud Foundations (Sep 2024)

Washington, DC

Jan 2024 – Dec 2025

St. Xavier's College

BS Computer Science & IT — **GPA: 3.89/4.0**

Kathmandu, Nepal

Aug 2018 – Dec 2022

Experience

George Washington University

Graduate Teaching Assistant – Design and Analysis of Algorithms

Washington, DC

Aug 2024 – Dec 2025

- Taught **60+ graduate students** algorithm concepts (dynamic programming, graph theory, NP-completeness) across **4 sections**, contributing to **22% improvement** in average midterm scores.
- Collaborated with faculty to develop standardized training materials for **8 TAs**, reducing onboarding time by **75%** while maintaining **95%+ grading consistency**.
- Redesigned curriculum with **3 faculty members** to incorporate real-world optimization problems, better preparing students for technical interviews and industry roles.

ECS Tech

Software Development Intern

Lalitpur, Nepal

Jun 2023 – Sep 2023

- Delivered **7 production features** for fintech client's payment platform using **React.js, Node.js, and PostgreSQL**, including automated booking system that eliminated **89% of scheduling conflicts** and saved **15 hours weekly**.
- Owned development and deployment of booking calendar, managing **\$5K budget** and coordinating with **5 stakeholders** to deliver feature **2 weeks ahead of schedule**, directly supporting client's **23% retention improvement**.
- Mentored **2 junior interns** on React optimization, achieving **43% reduction** in page load times that increased user conversion from **12% to 16%**.

Intel Security Service Pvt. Ltd.

IT Solutions Lead (Software & Web Systems) – Contract

Kathmandu, Nepal

Aug 2022 – Mar 2023

- Led digital transformation for **25-person** consultancy, rebuilding website that increased qualified leads by **73%** and generated **\$35K in new contracts**.
- Reduced IT operational costs by **40%** while managing **\$12K annual budget** and eliminating **\$35K external contractor dependency** through in-house solutions.

Projects

InfraSight – Cloud Cost Intelligence Platform — Live Demo — GitHub

2025

Python, FastAPI, PostgreSQL, scikit-learn, Docker, React, AWS Cost Explorer

- Built cloud cost intelligence platform with **92% ML prediction accuracy**, reducing AWS billing analysis time by **70%** for DevOps teams managing **\$10K+ monthly cloud spend**.
- Implemented ML forecasting pipelines (**Random Forest + LSTM, Z-score analysis**) enabling teams to catch cost spikes before billing cycles close.
- Architected production-ready backend with **Docker, PostgreSQL, JWT authentication**, and **sub-200ms inference time**.

MelodyHub – Real-Time Social Music Platform — GitHub

2025

Node.js, React, Socket.IO, Clerk Auth, Cloudinary, MongoDB

- Built real-time streaming platform supporting **100+ concurrent users** with synchronized playback and live chat, achieving **99%+ uptime** during 2-month testing with **50+ GW students**.
- Led **3-person team** using Agile, creating OOP backend architecture that reduced feature development time by **60%**.
- Integrated **Cloudinary CDN** and **Clerk authentication**, handling **500MB+ daily uploads** while maintaining **sub-3-second load times**.

TalkifyDocs – AI Document Chat Application — Live Demo — GitHub

2023

Next.js, OpenAI GPT-4, LangChain, Pinecone, Stripe, Prisma, TypeScript

- Built AI-powered document chat SaaS demonstrating **RAG architecture**, implementing authentication, **Stripe payments**, and subscription billing within **\$150 monthly API budget**.
- Achieved **sub-200ms query responses** on **10GB+ repositories** with **94% parsing accuracy** using **LangChain** contextual memory and **Pinecone** vector database for semantic search.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

Frontend: React.js, Next.js, Tailwind CSS, Flutter

Backend & APIs: Node.js, FastAPI, Flask, RESTful Services

Databases: PostgreSQL, MongoDB, MySQL

Cloud & DevOps: AWS (Cost Explorer, EC2, S3, IAM), Docker, Git, Linux, CI/CD

ML/AI: scikit-learn, TensorFlow, Keras, LangChain, FAISS, Pinecone, RAG

Tools: Agile/Scrum, Unit Testing, Data Structures & Algorithms, OOP, System Design